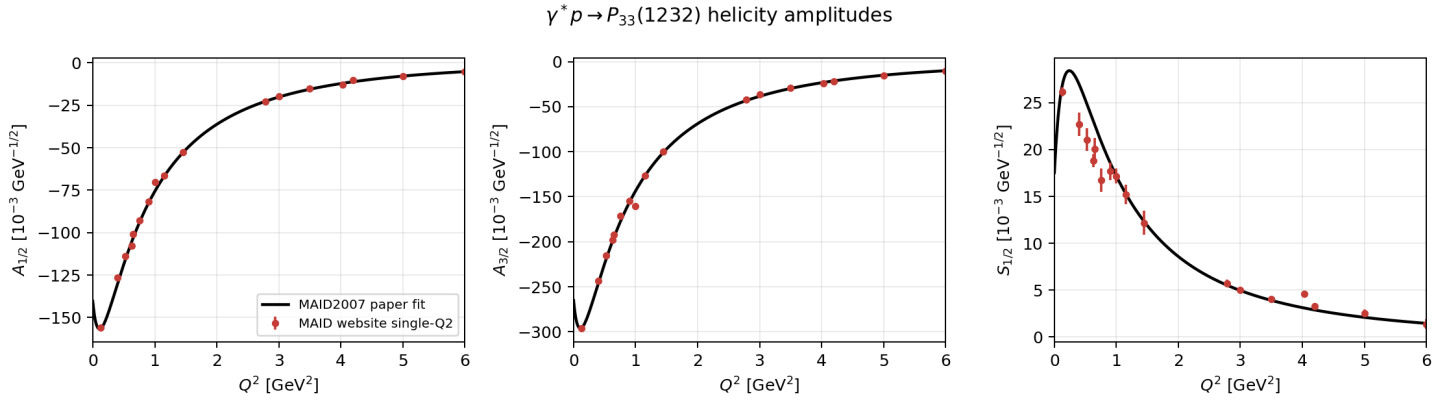


MAID2007 GENIE Check Plots - Corrected Final Set

Final/corrected validation plots only; intermediate plots with known issues are excluded.

P33 helicity amplitude inputs

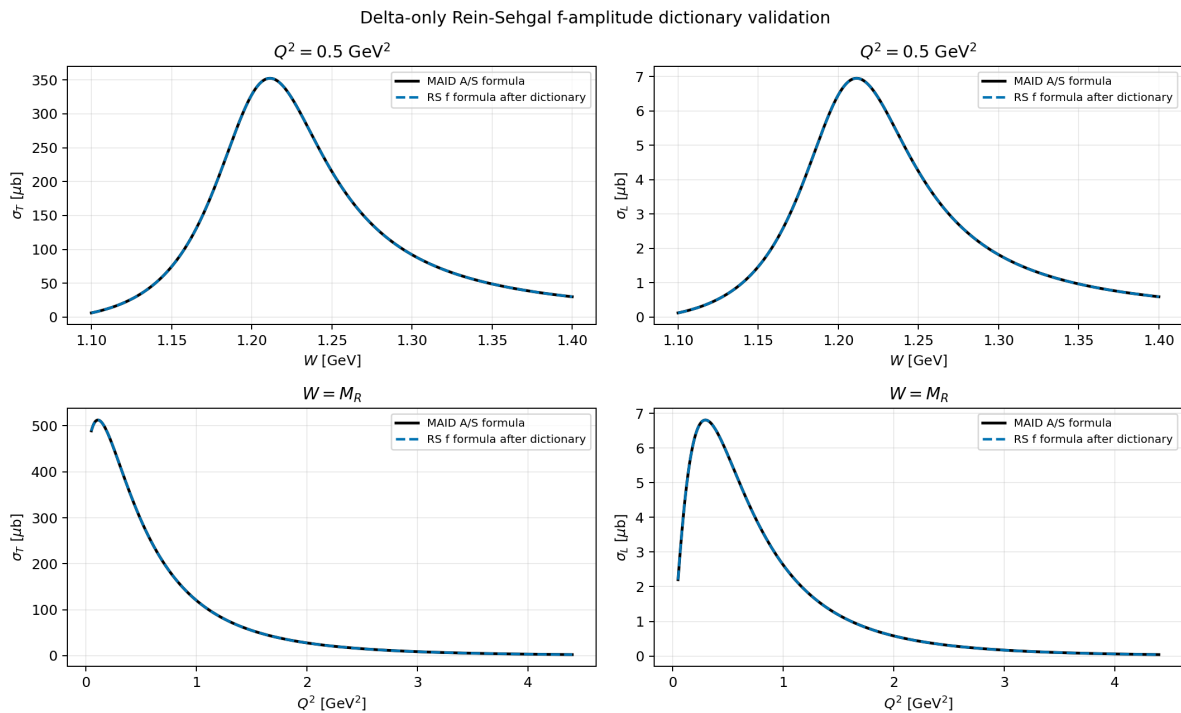
MAID2007 P33 input amplitudes against source points.



Source: validation/maid_form_factor_validation/figures/maid2007_p33_helicity_inputs.png

Delta-only RS-MAID dictionary validation

A/S-to-f dictionary check for the Delta route.



Source: validation/maid_form_factor_validation/figures/rs_maid_dictionary_validation.png

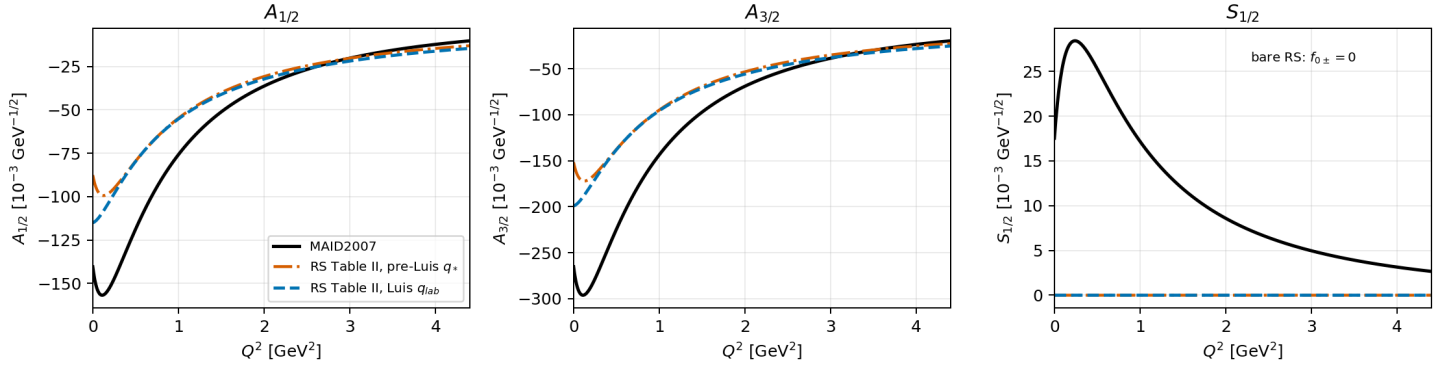
MAID2007 GENIE Check Plots - Corrected Final Set

Final/corrected validation plots only; intermediate plots with known issues are excluded.

Original RS P33 amplitudes converted to A/S

Reference comparison of original RS amplitudes and MAID A/S conventions.

Original Rein-Sehgal P33 f amplitudes converted to A/S

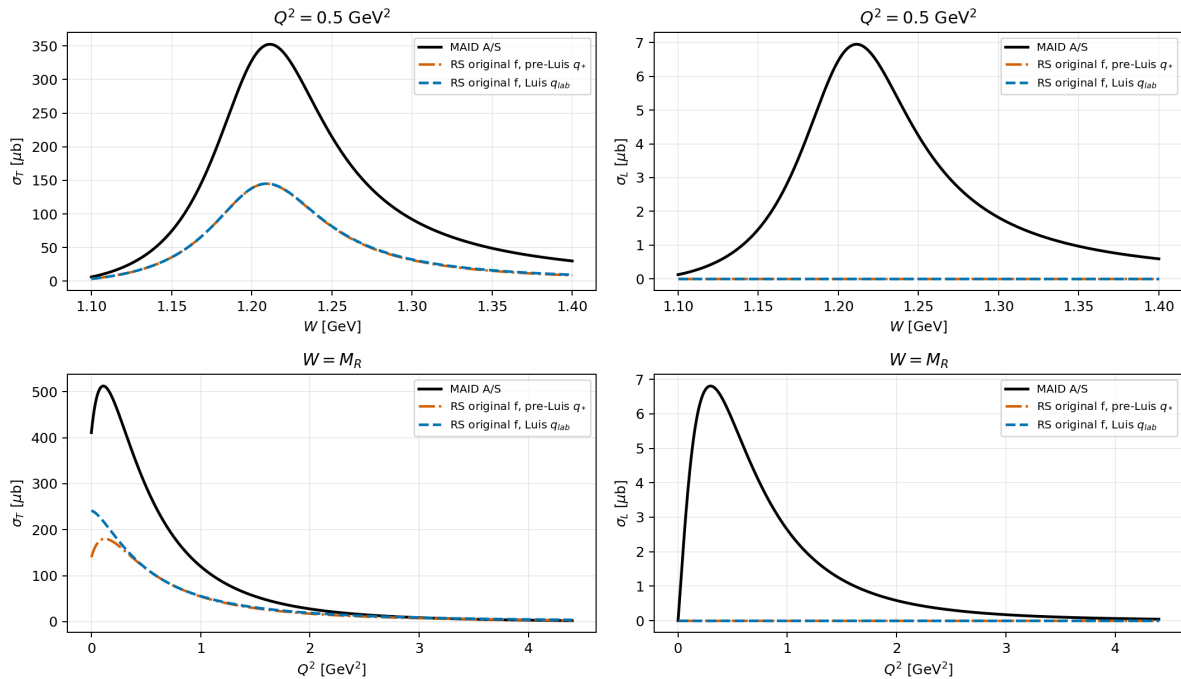


Source: validation/maid_form_factor_validation/figures/rs_original_p33_as_vs_maid.png

Original RS P33 cross sections versus MAID

Reference cross-section comparison for the original RS P33 form factors.

Original Rein-Sehgal P33 f-amplitude cross sections vs MAID



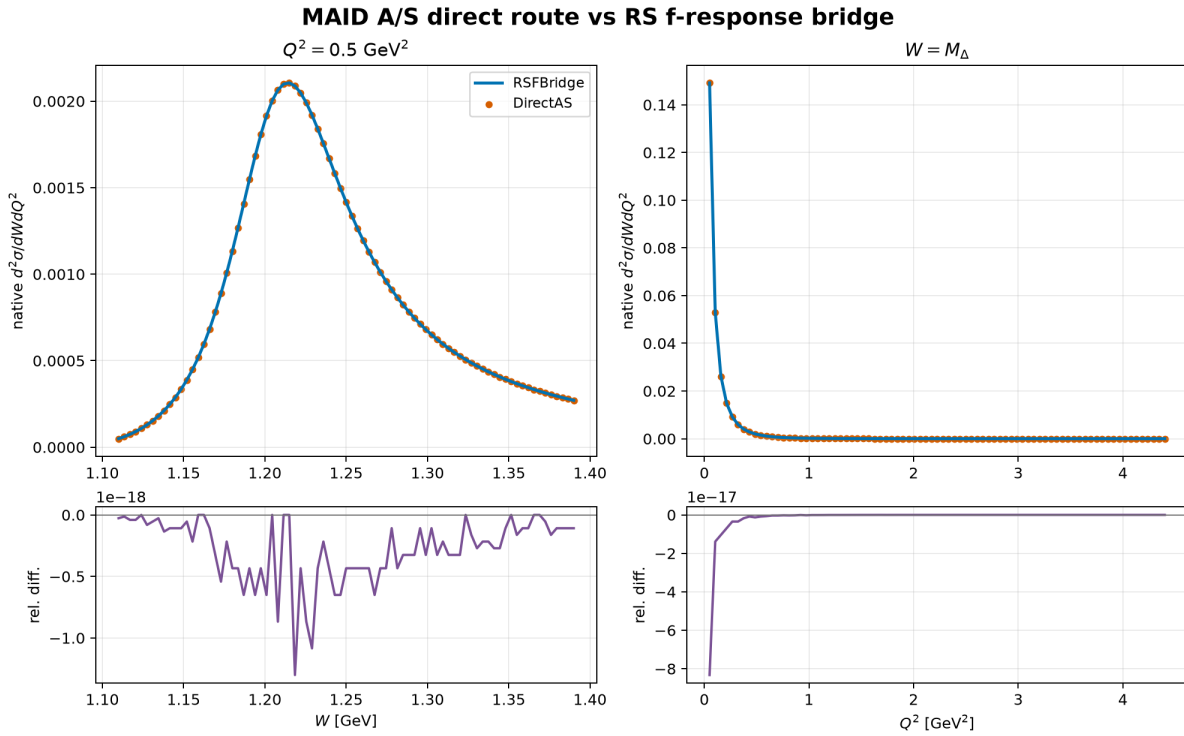
Source: validation/maid_form_factor_validation/figures/rs_original_p33_cross_sections_vs_maid.png

MAID2007 GENIE Check Plots - Corrected Final Set

Final/corrected validation plots only; intermediate plots with known issues are excluded.

GENIE MAID Delta bridge route comparison

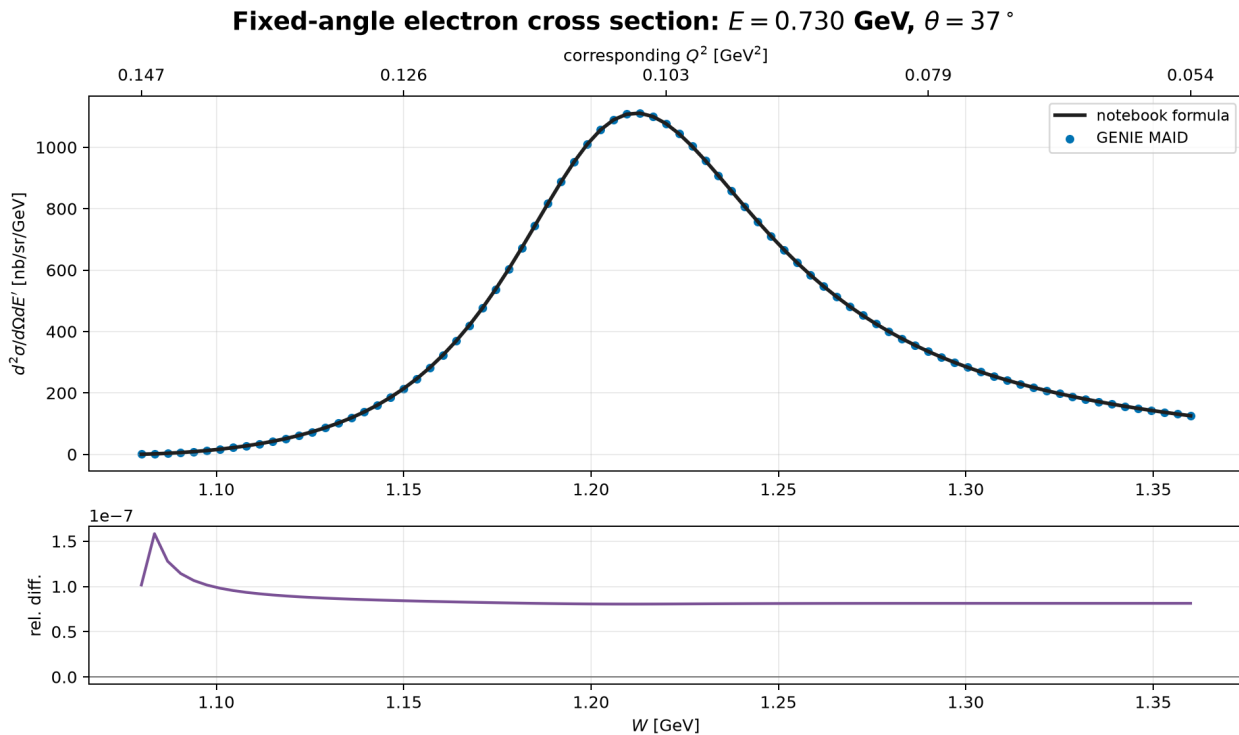
Direct A/S route versus RS f-response bridge.



Source: validation/maid_form_factor_validation/figures/genie_maid_delta_bridge_route_comparison.png

GENIE MAID Delta fixed-angle cross section

Fixed-angle Delta comparison to the notebook formula.



Source: validation/maid_form_factor_validation/figures/genie_maid_delta_fixed_angle_comparison.png

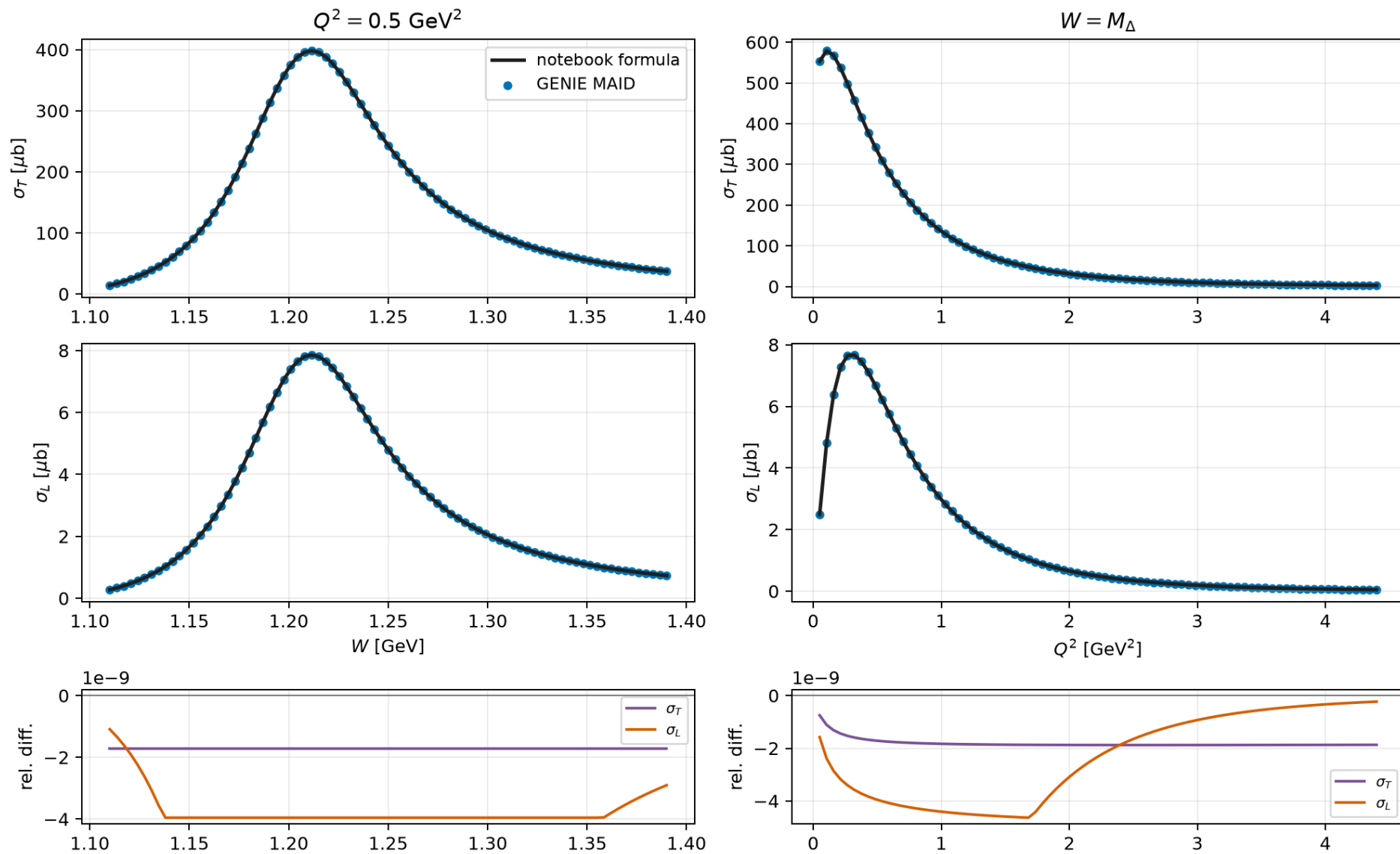
MAID2007 GENIE Check Plots - Corrected Final Set

Final/corrected validation plots only; intermediate plots with known issues are excluded.

GENIE MAID Delta absorption cross sections

Delta-only absorption cross-section validation.

Delta-only MAID absorption cross sections



Source: validation/maid_form_factor_validation/figures/genie_maid_delta_absorption_comparison.png

MAID2007 GENIE Check Plots - Corrected Final Set

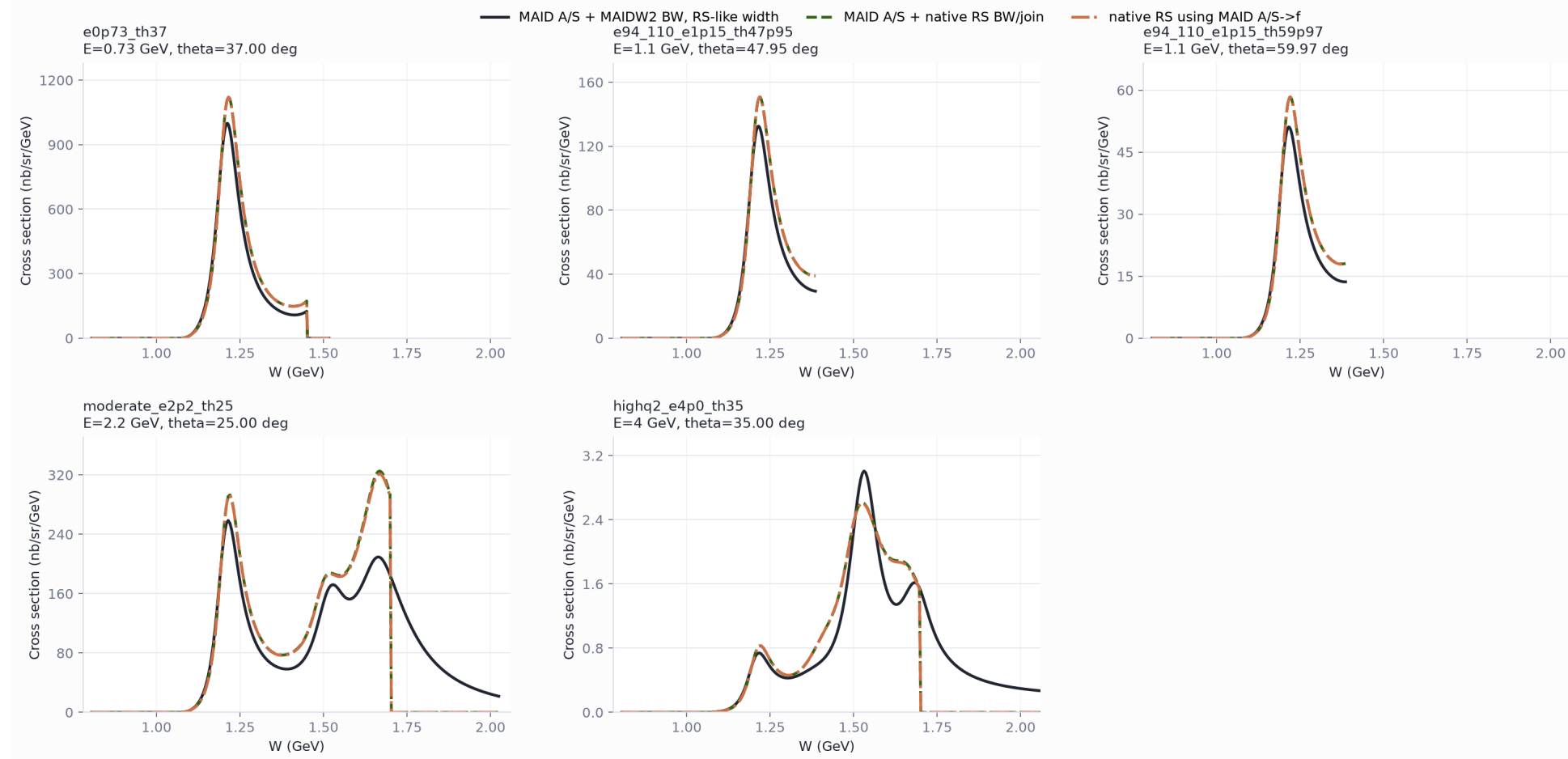
Final/corrected validation plots only; intermediate plots with known issues are excluded.

MAID2007 amplitudes with MAIDW2 versus native RS Breit-Wigner

Corrected no- π N-branching, RS-like width comparison.

MAID2007 amplitudes: MAIDW2 versus native RS Breit-Wigner

Free proton; fixed-angle electron kinematics; y is nb/sr/GeV after the GENIE kPSWQ2fE to fixed-angle Jacobian.



Source: validation/maid_allres_compare/out/plots/matplotlib/maid_vs_rs_maidf_overlays_mpl.png

MAID2007 GENIE Check Plots - Corrected Final Set

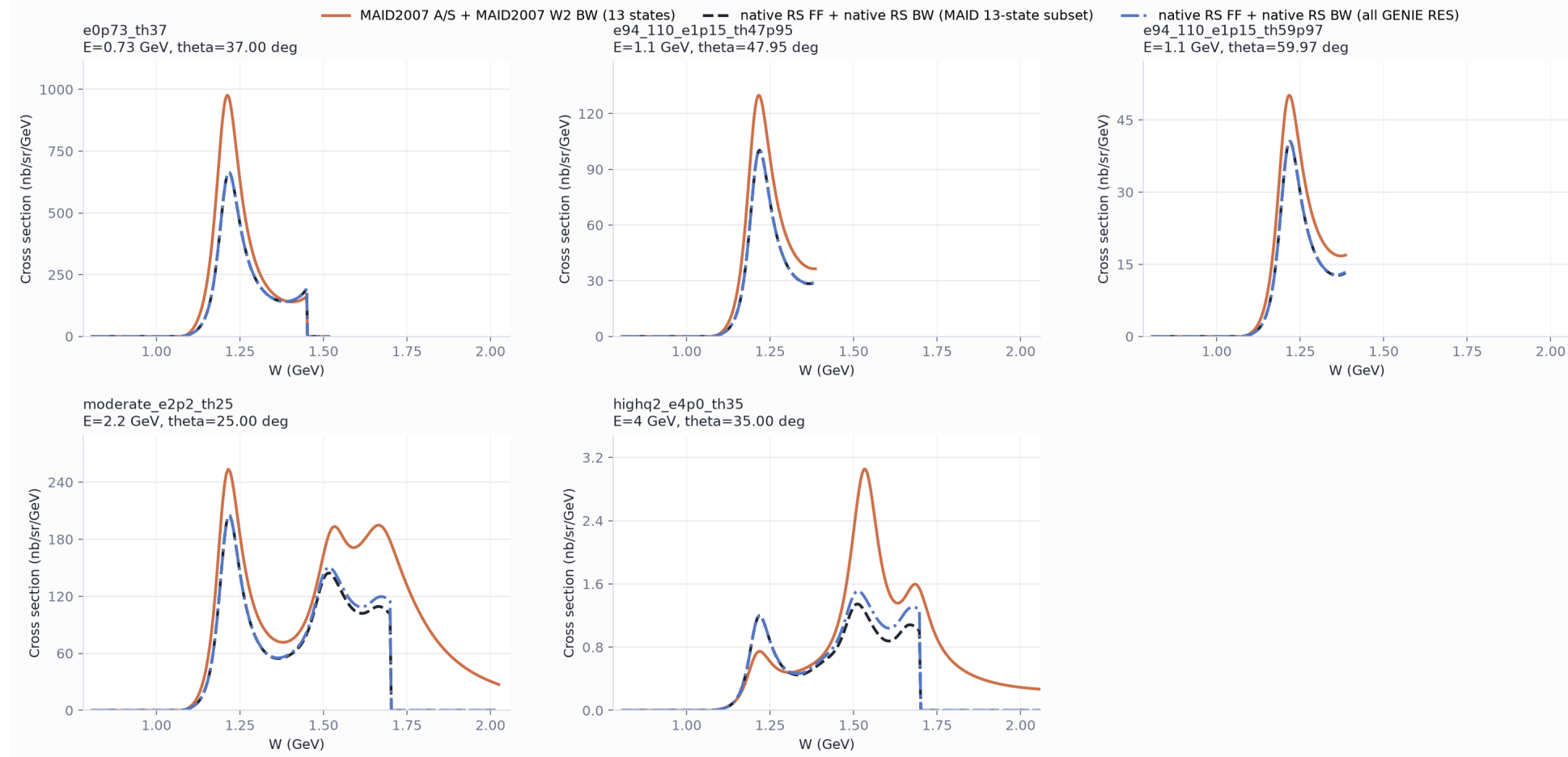
Final/corrected validation plots only; intermediate plots with known issues are excluded.

Native RS versus MAID2007 formalism comparison

Each formalism uses its own form-factor and Breit-Wigner conventions.

Native RS versus MAID2007 with each formalism's FF and BW

Free proton; fixed-angle electron kinematics; MAID uses MAID2007 A/S form factors and MAID2007 W2 BW, no pi-N branch.



Source: validation/maid_allres_compare/out/plots/matplotlib/formalism_native_rs_vs_maid_mpl.png

MAID2007 GENIE Check Plots - Corrected Final Set

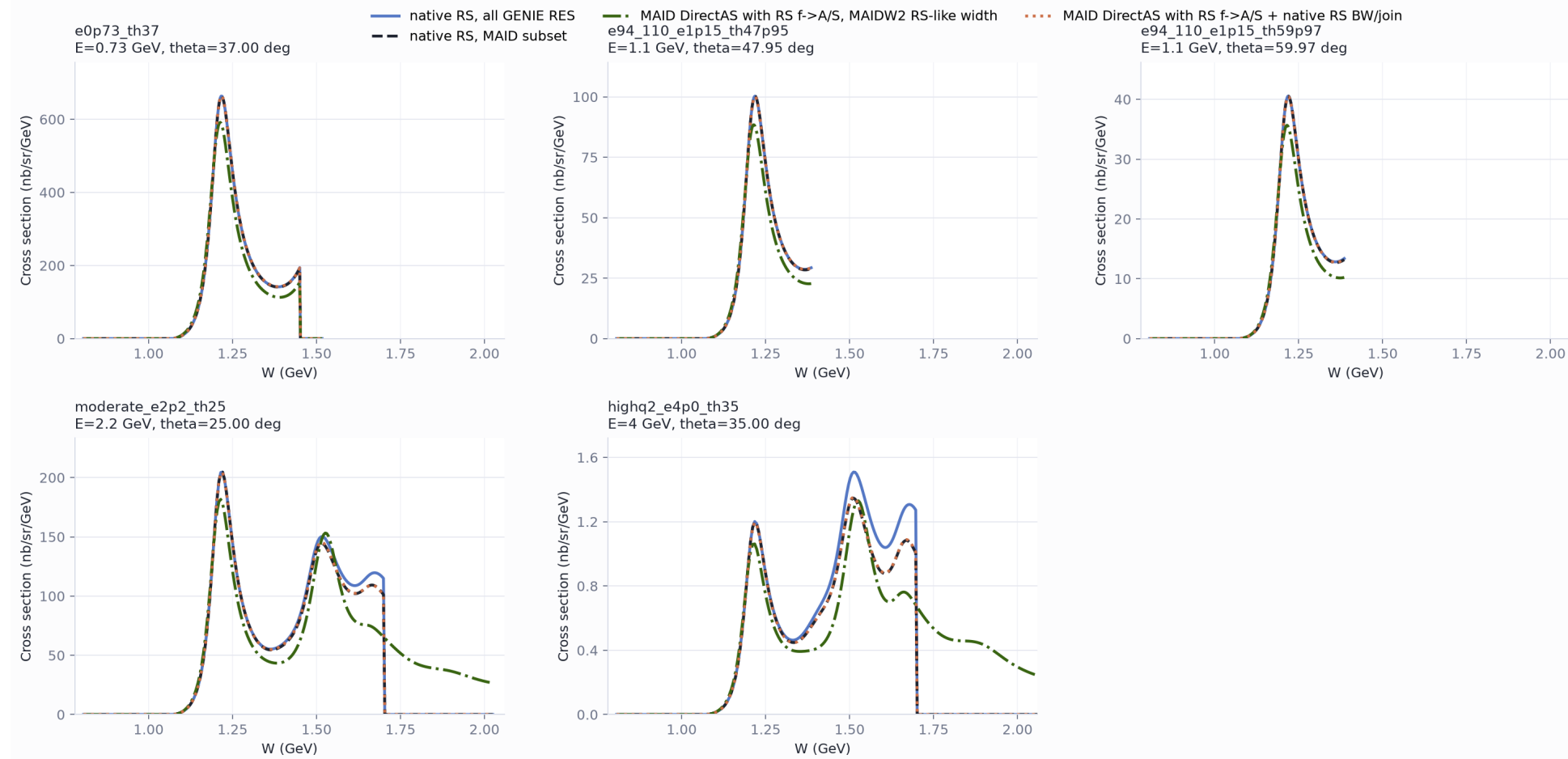
Final/corrected validation plots only; intermediate plots with known issues are excluded.

Native RS full set, MAID subset, and MAID A/S-to-f comparison

Native RS all resonances versus MAID 13-state subset and MAID-derived f amplitudes.

Native RS resonance set versus MAID-subset controls

Free proton; fixed-angle electron kinematics; MAID-subset curves use the 13 MAID2007 resonances only.



Source: validation/maid_allres_compare/out/plots/matplotlib/rs_full_subset_and_maid_rs_overlays_mpl.png

MAID2007 GENIE Check Plots - Corrected Final Set

Final/corrected validation plots only; intermediate plots with known issues are excluded.

Integrated cross-section ratios

Ratios over the fixed-angle W scans for the BW/amplitude comparison.

Integrated cross-section ratios

Trapezoidal integrals over each W grid; free proton; default MAIDW2 uses RS-like running width and no pi-N branching.



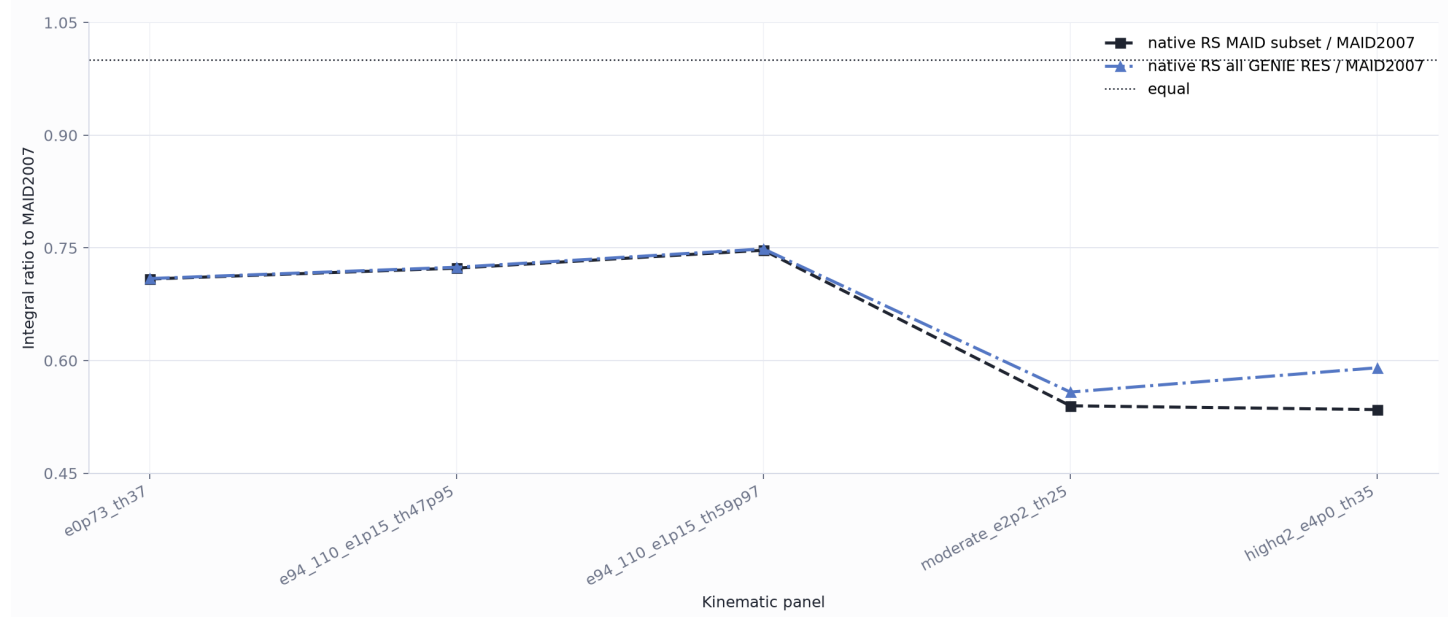
Source: validation/maid_allres_compare/out/plots/matplotlib/integral_ratio_summary_mpl.png

Formalism integrated ratios

Native RS subset/all-resonance integrated ratios relative to MAID2007.

Formalism comparison: integrated native RS relative to MAID2007

Trapezoidal integrals over each W grid; MAID uses MAID2007 A/S form factors and MAID2007 BW convention.



Source: validation/maid_allres_compare/out/plots/matplotlib/formalism_integral_ratios_mpl.png

MAID2007 GENIE Check Plots - Corrected Final Set

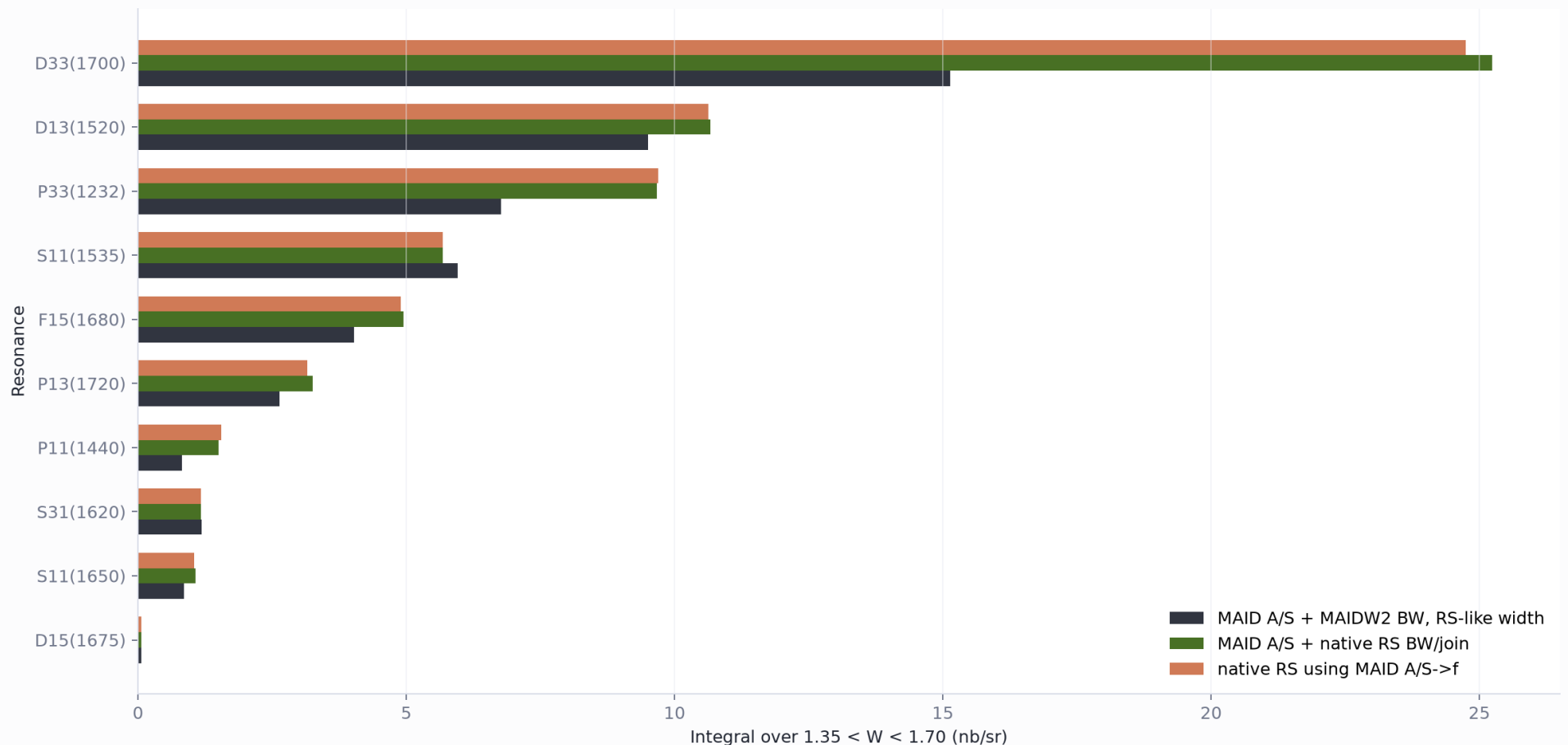
Final/corrected validation plots only; intermediate plots with known issues are excluded.

Per-resonance Breit-Wigner audit

Moderate E=2.2 GeV, theta=25 deg by-resonance integral audit.

Per-resonance Breit-Wigner audit

moderate_e2p2_th25; integrals compare default MAIDW2 width, native RS BW, and native RS using MAID A/S->f.



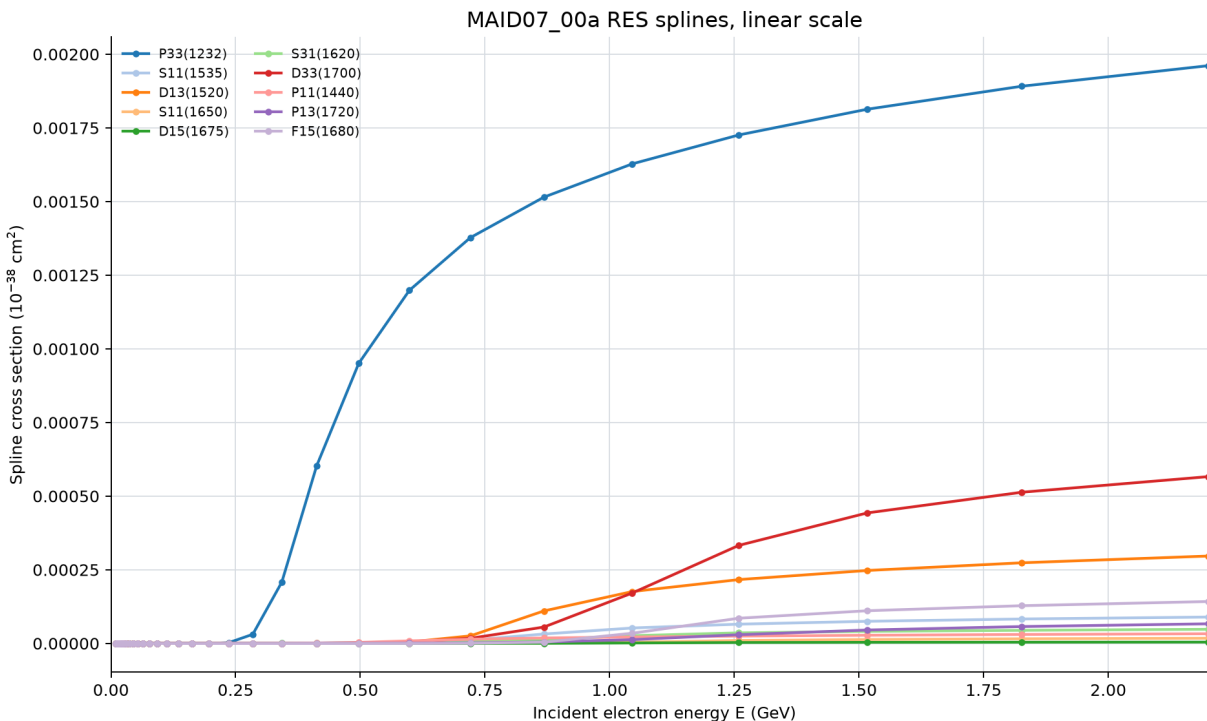
Source: validation/maid_allres_compare/out/plots/matplotlib/moderate_e2p2_th25_by_resonance_integrals_mpl.png

MAID2007 GENIE Check Plots - Corrected Final Set

Final/corrected validation plots only; intermediate plots with known issues are excluded.

MAID07_00a RES splines - linear scale

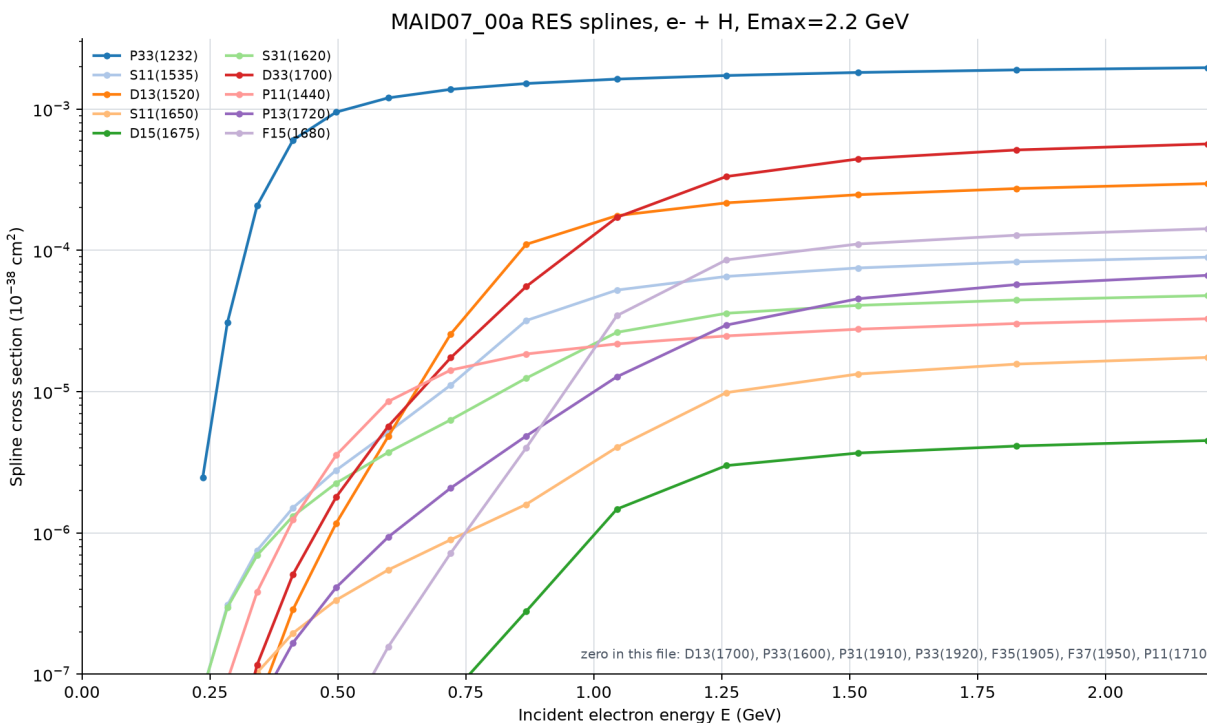
Dedicated MAID integrator spline check.



Source: validation/maid_splines/out/plots/maid07_res_splines_linear.png

MAID07_00a RES splines - log scale

Dedicated MAID integrator spline check, log scale.



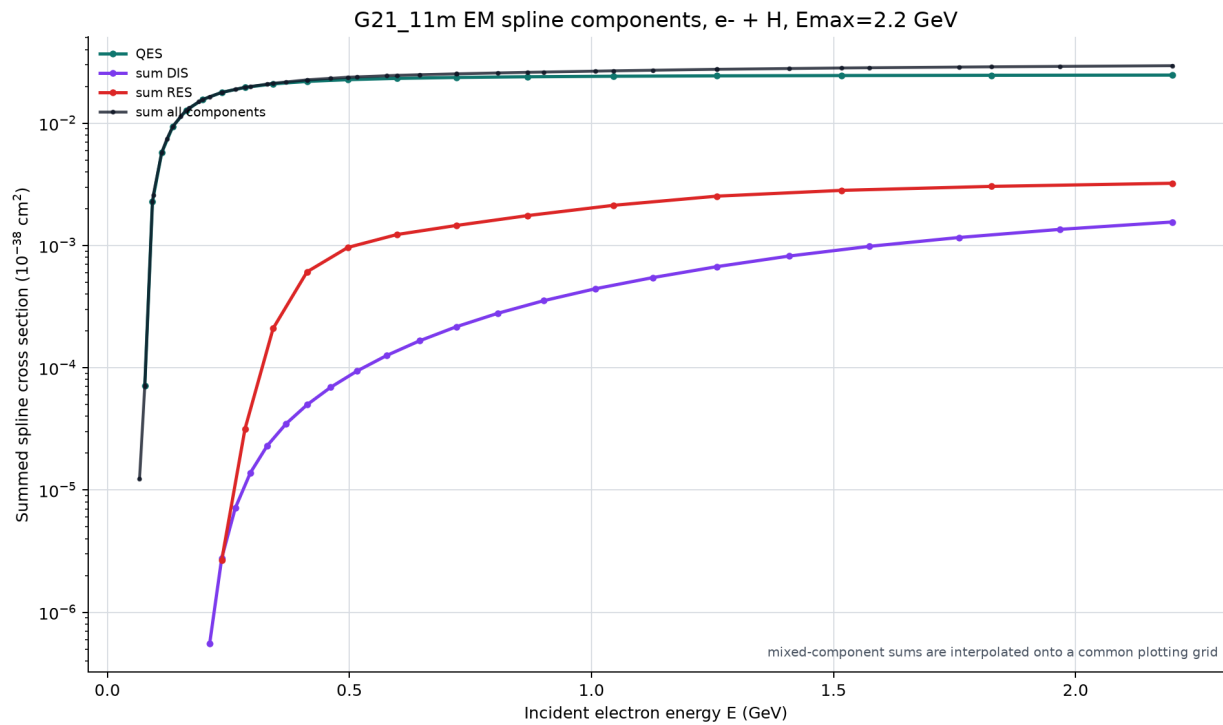
Source: validation/maid_splines/out/plots/maid07_res_splines_log.png

MAID2007 GENIE Check Plots - Corrected Final Set

Final/corrected validation plots only; intermediate plots with known issues are excluded.

G21_11m EM component splines

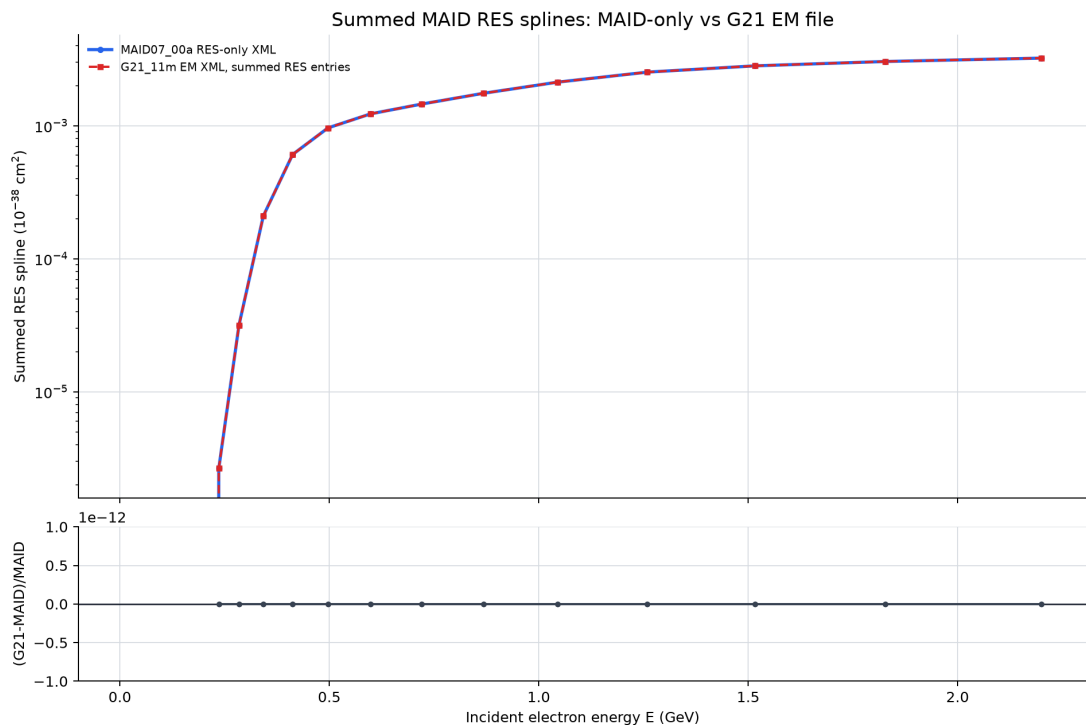
G21 EM variant component spline check.



Source: validation/maid_splines/out/plots/g21_em_component_splines.png

MAID-only RES spline versus G21 EM RES sum

RES part extracted from full G21 EM spline agrees with MAID-only RES spline.



Source: validation/maid_splines/out/plots/maid_vs_g21_res_sum_overlay.png

MAID2007 GENIE Check Plots - Corrected Final Set

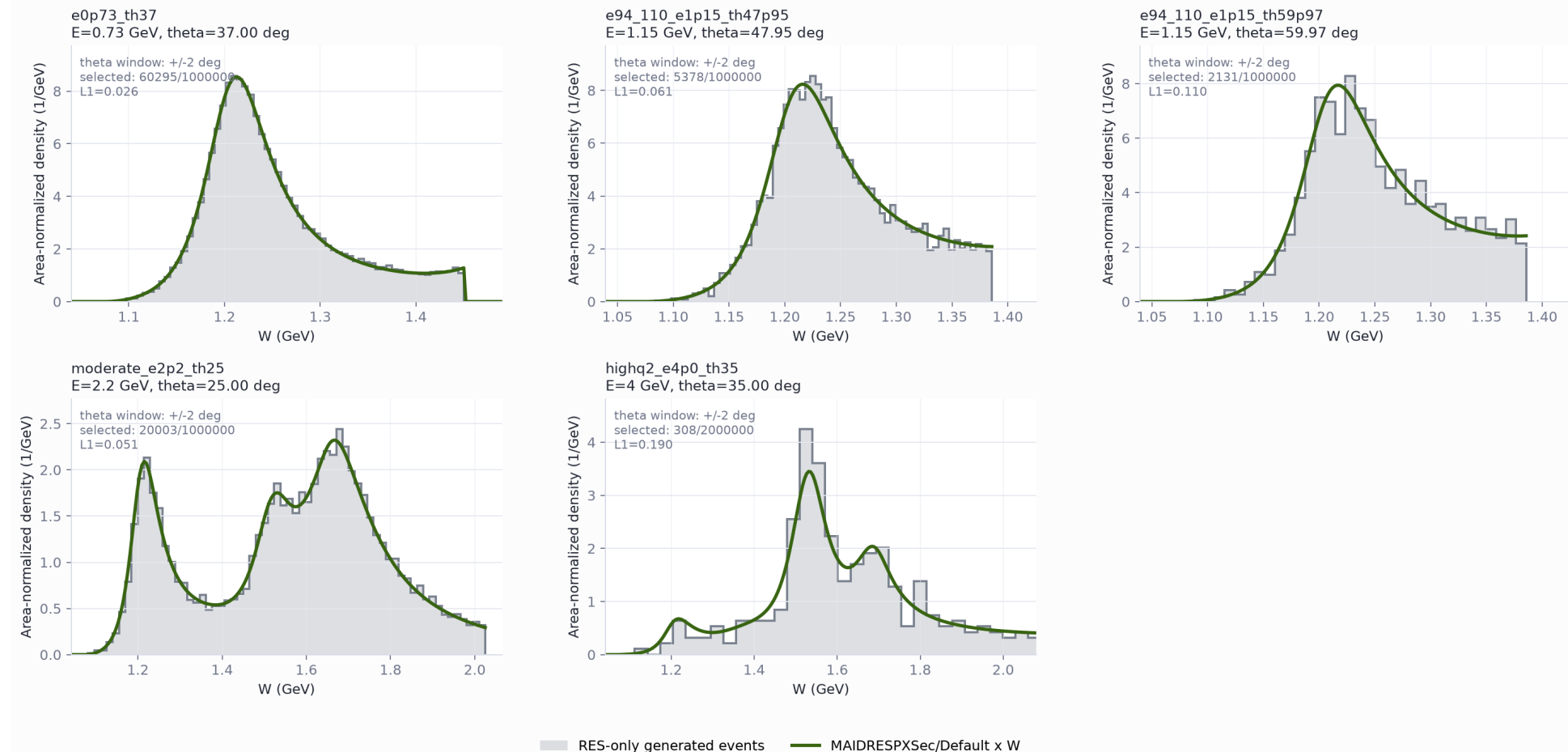
Final/corrected validation plots only; intermediate plots with known issues are excluded.

RES-only generated events versus MAIDRESPXSec/Default, theta window ± 2 deg

Clean final event comparison with MAIDRESPXSec/Default and the corrected xW factor.

RES-only generation versus MAIDRESPXSec/Default

Free proton; fixed-angle curve multiplied by W for comparison to theta-sliced event W densities. All shapes are area-normalized.



Source: validation/maid_splines/out/res_only_angle_slice_events/plots/res_only_events_vs_maid_default_xW_clean_eps2p0.png

MAID2007 GENIE Check Plots - Corrected Final Set

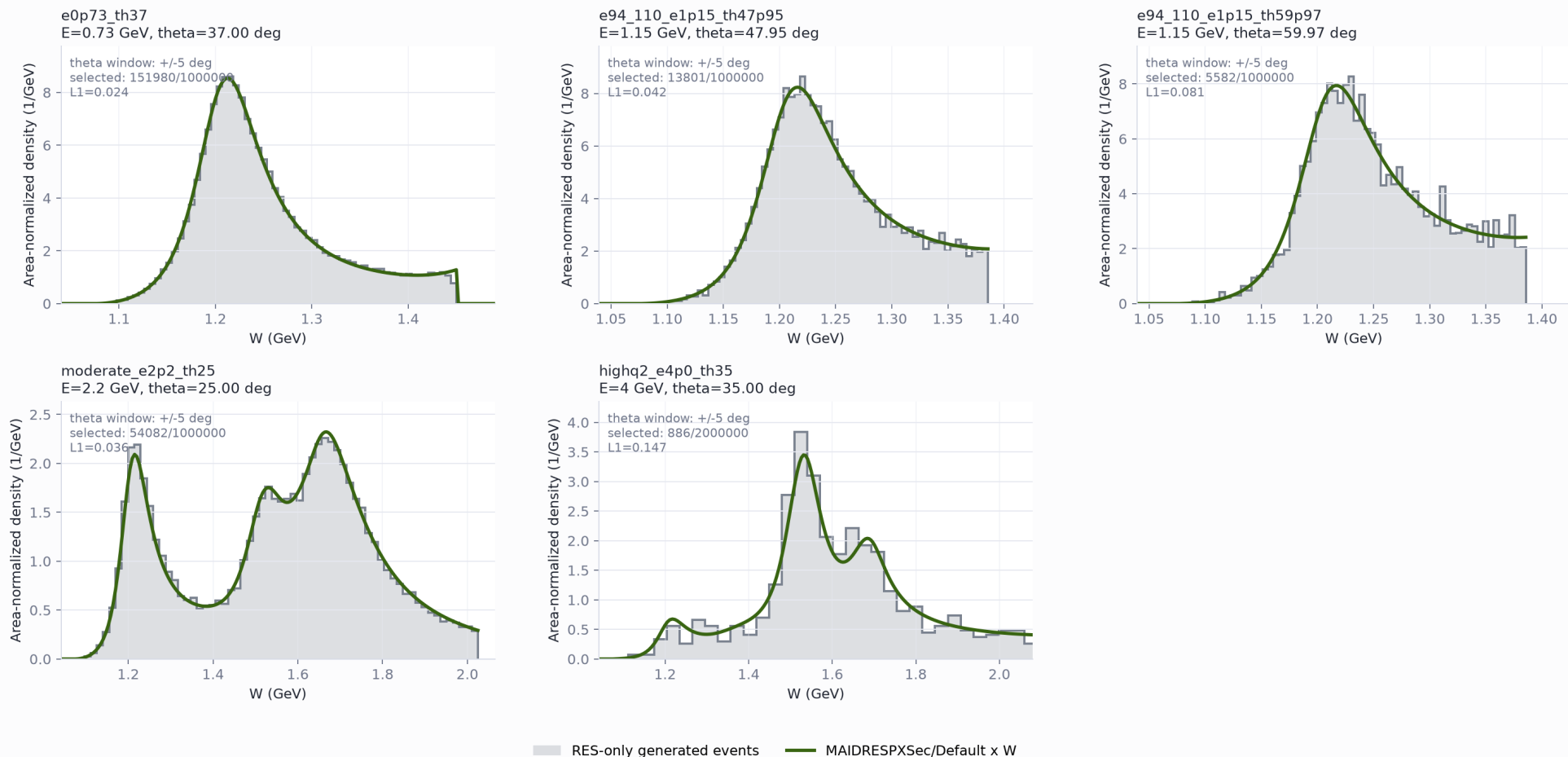
Final/corrected validation plots only; intermediate plots with known issues are excluded.

RES-only generated events versus MAIDRESPXSec/Default, theta window +/- 5 deg

Clean final event comparison with MAIDRESPXSec/Default and the corrected xW factor.

RES-only generation versus MAIDRESPXSec/Default

Free proton; fixed-angle curve multiplied by W for comparison to theta-sliced event W densities. All shapes are area-normalized.



Source: validation/maid_splines/out/res_only_angle_slice_events/plots/res_only_events_vs_maid_default_xW_clean_eps5p0.png